General SPT Stability Rules

MR Troubleshooting Seminar

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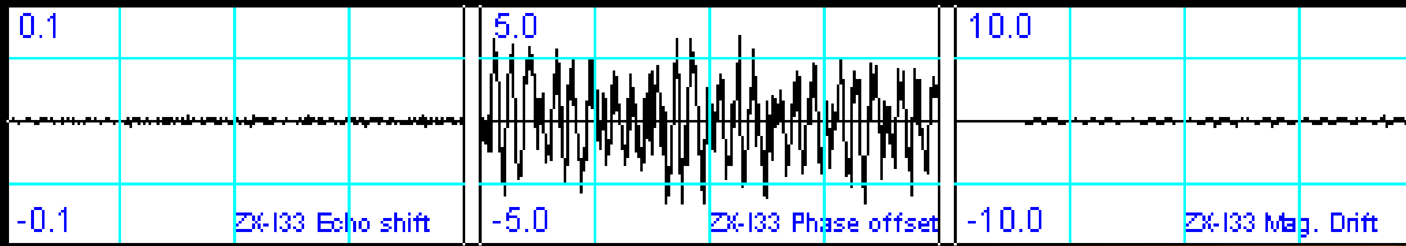
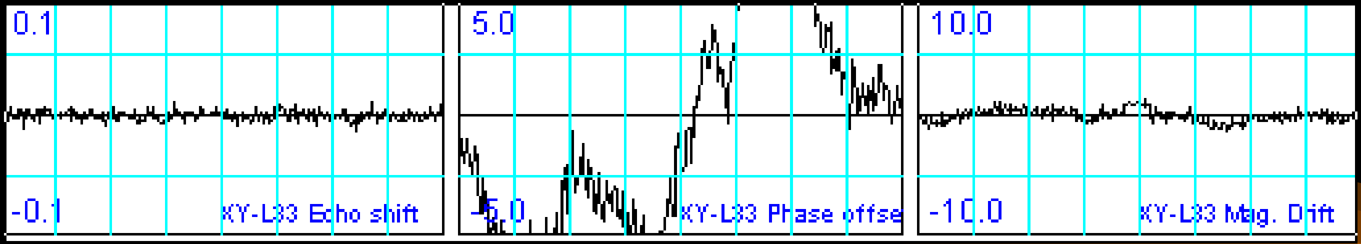
Buc - June, 2001

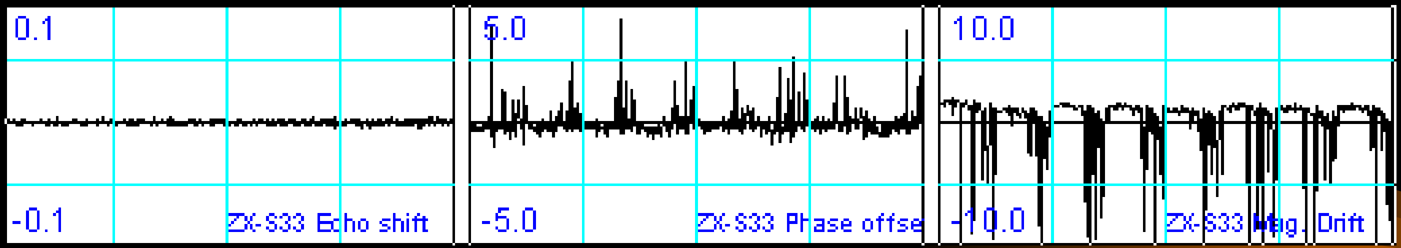


- **If FSE**
- **High Phase**
- No Echo Center
- No Magnitude

- Then
- **Magnetic Problem**
- (e.g. vibration – turn off coldhead, air conditioners, etc.
- may need to use HSS to isolate a vibration problem)



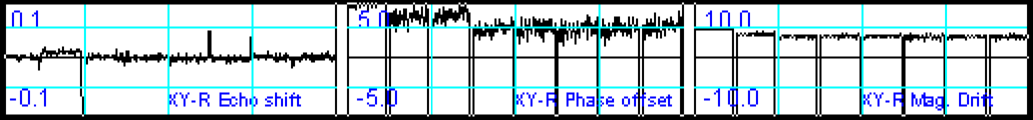
- **If FGRE**
- **High Phase**
- No Echo Center
- No Magnitude
- Then
- **Magnetic Problem**
- (Moving Metal (truck, train, elevator, etc) or EMI (i.e. train)

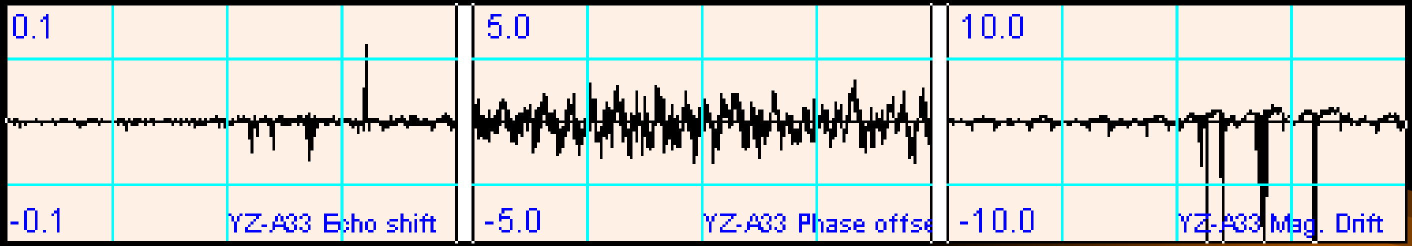


- **If FSE or GRE**
- **High Phase <u>and</u> Magnitude**

- Then
- **Transmit Fault**
- (RF amp. Or anything else in xmit chain (dd, tr, coil, etc))

Sometimes effects Echo-center also





- **If FSE or FGRE has**
- **Magnitude only**

Sometimes effects Echo-center also

- Then
- **Receive chain fault**
- (preamp, TNS, or anything else in the rec. chain)

Note: FSE Receive faults do NOT propagate down the echo-train (FSE xmit faults do)

Eddy Currents

- Eddy currents effects can sometimes be seen in the FGRE plots.
- Use the Body plots for this analysis since they have a center-slice and are further from iso-center.
- If - the eddy current is detected in Echo-center plots

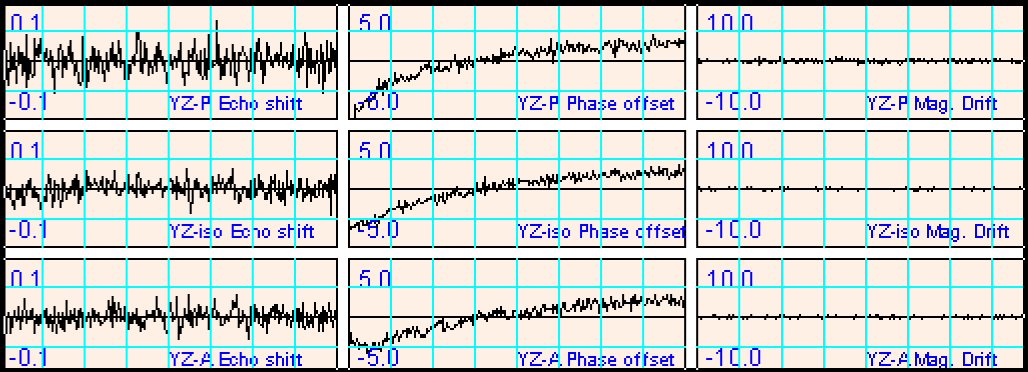
Then it's coming from the Readout Gradient

- If - the eddy current is detected in the Phase plots

Then it's coming from the Slice Select gradient

Eddy Currents (b0)

- **If eddy current on Slice-select**
- **Are all in the same direction (polarity)**
- Then
- **It's a B0 eddy current**

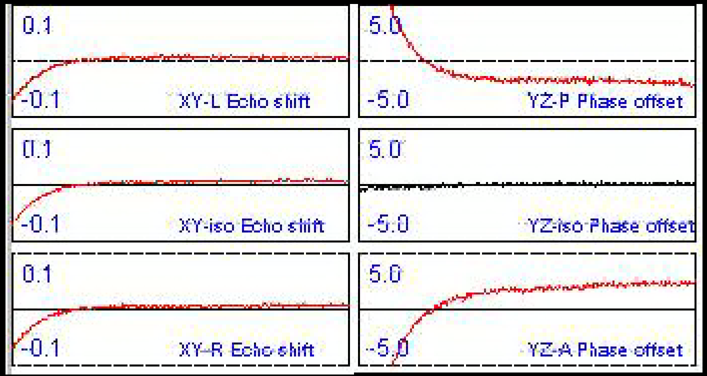


Eddy Currents (linear)

- **If eddy current on Slice-select**
- **Change polarity**

- Then
- **It's a linear eddy current**

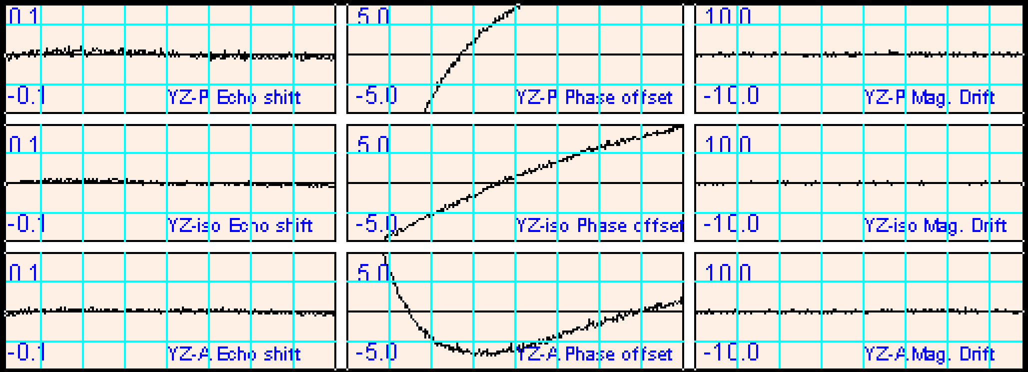
Can't tell what type of eddy current, if looking at Echo shift



Can tell what type of eddy current, when looking at Phase offset

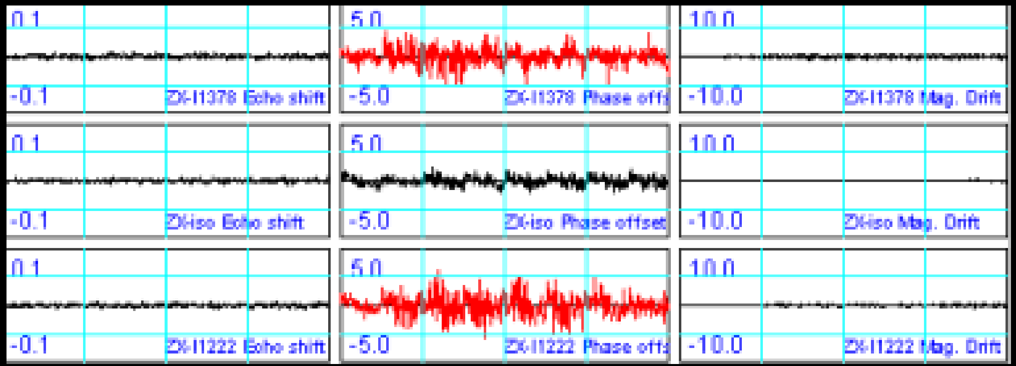
Eddy Currents (combo)

- **It's also possible to see a b0 and linear term if both are present**

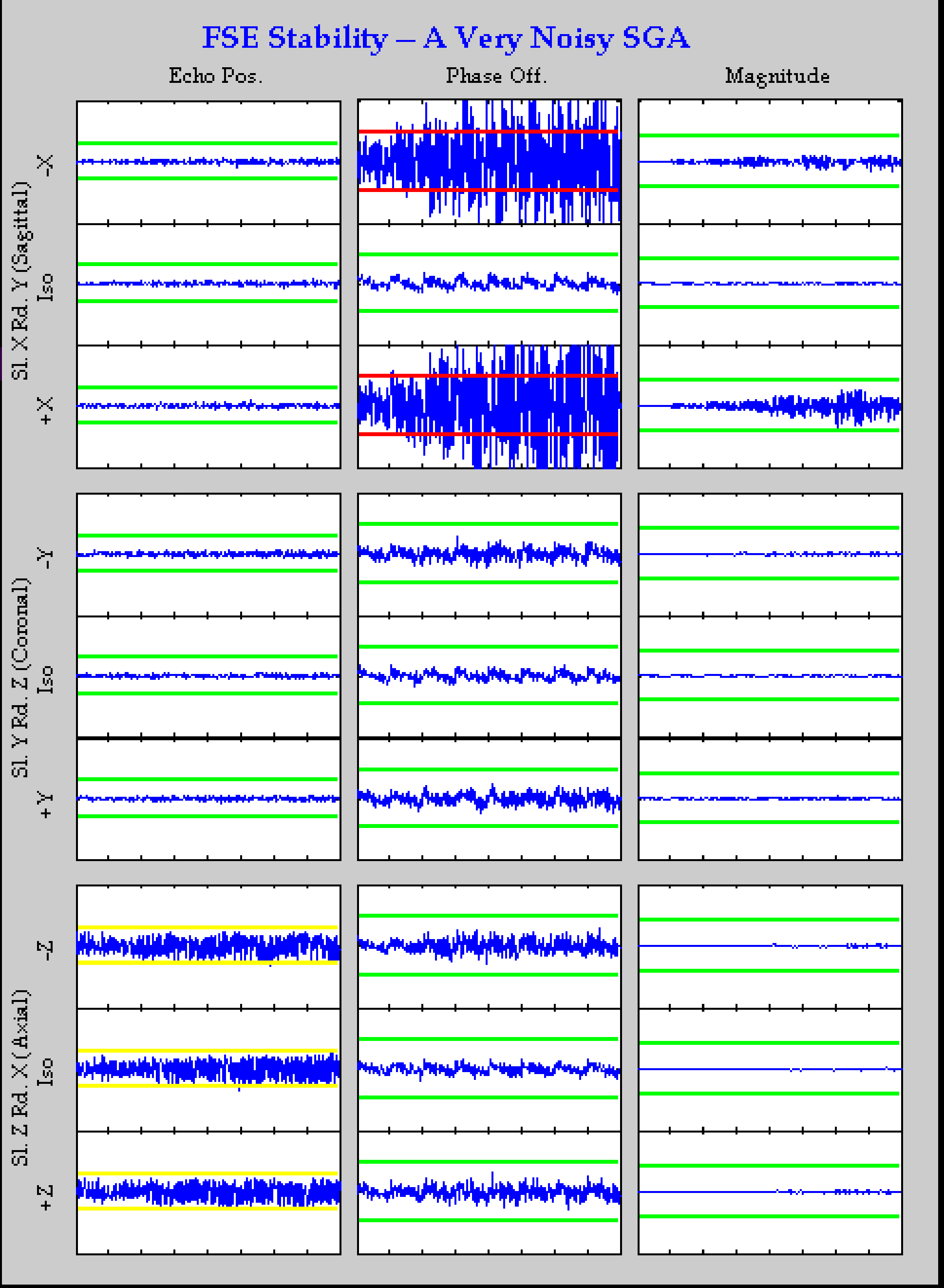


Gradient Signatures

- If it's the **Echo-Center** Plots that have a problem Then it's caused by the **Readout Gradient**
- If the **Phase plots look OK on the iso-center** plots, **and the off iso-center plots are bad**
- Then it's caused by the **Slice-Select** Gradient



X as Slice Select



X as Readout

Gradient Signatures

Intermittent steps in the waveforms. Based on rules in

pg. 10, points at the Z axis (this problem was in the GP, most likely – noisy DAC.

Body FSE

Body GRE

SS = X

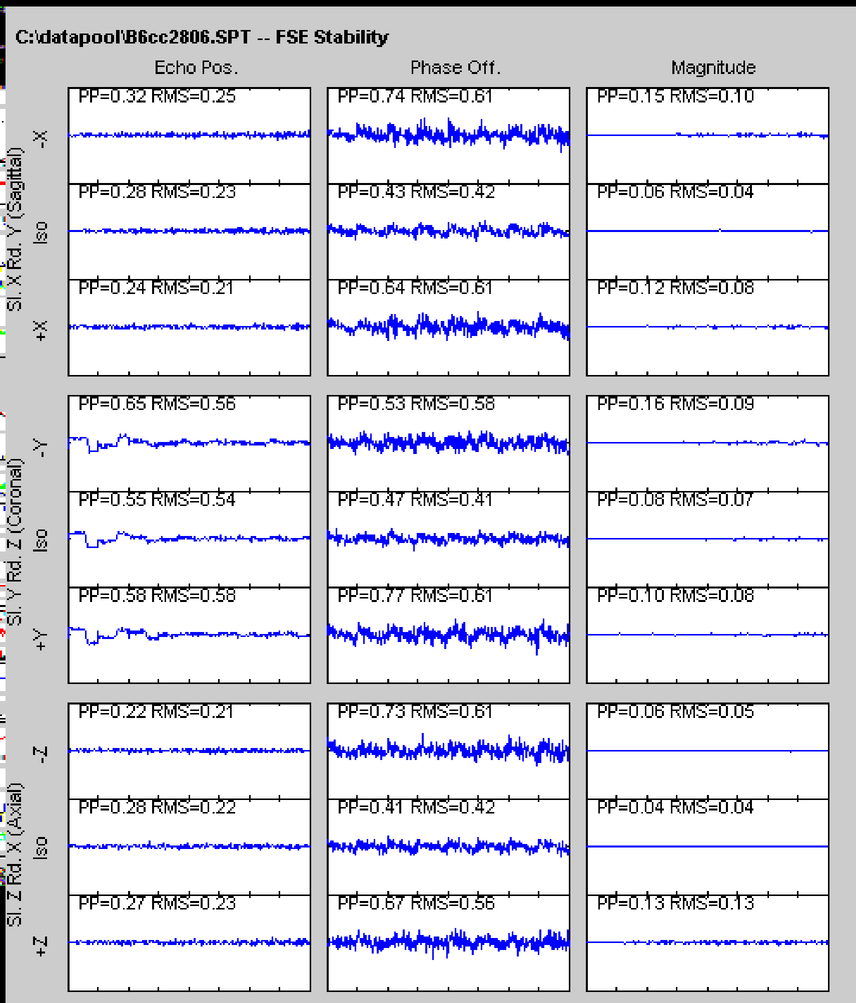
RO = Y

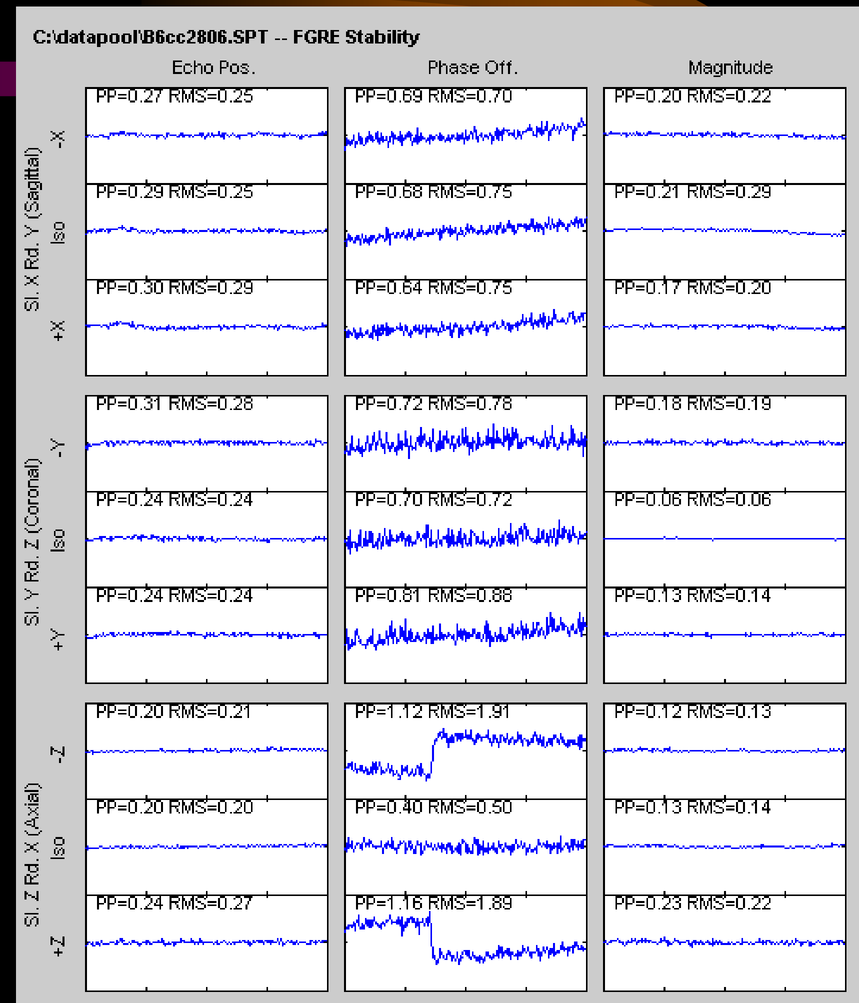
SS = Y

RO = Z

SS = Z

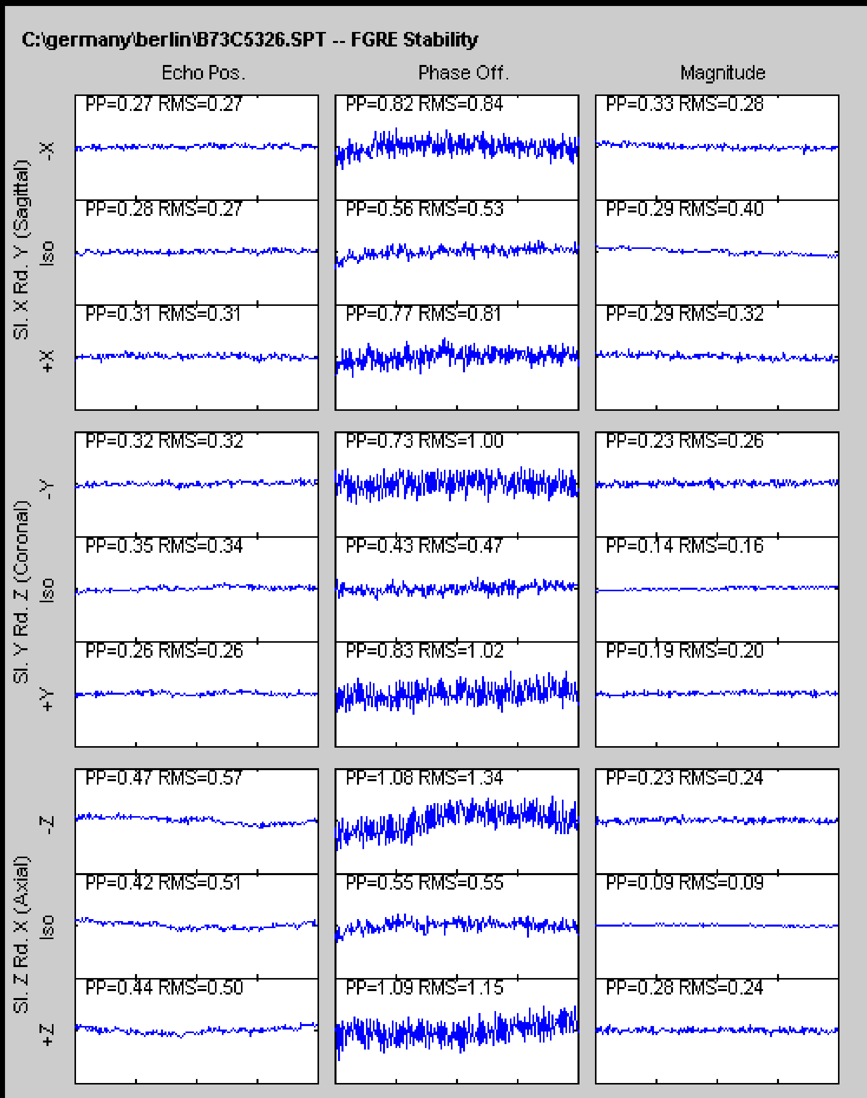
RO = X

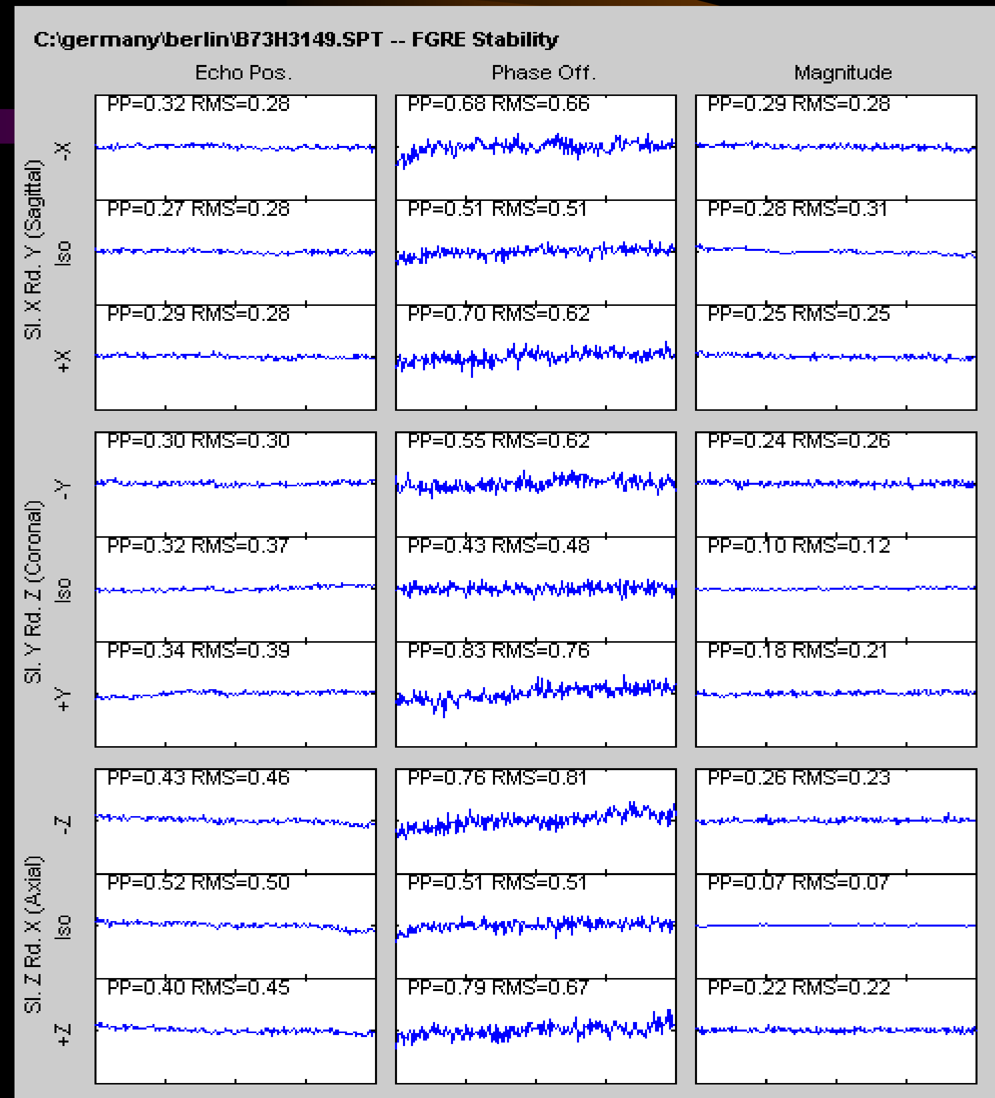




Gradient Signatures

Based on rule in pg. 10, points at the slice-select gradient (clean phase at iso-center, high phase off-iso). This was a bad ACGD Power Supply Control Bd.





Bad PS Cont. Bd.

New PS Cont. Bd.